

ReadTheWay: Designing and Developing a Mobile Application to Facilitate and Promote Literary Tourism

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Abstract

This work describes the design and development of ReadTheWay, a mobile application for cultural and literary tourism that enables travellers to connect with their destinations through stories, poems, and historical narratives. It addresses the lack of digital solutions for cultural tourism by placing literature at the centre of the user experience while offering interactive navigation. This paper describes a project that began with market research and interviews with potential users, leading to the development of a first high-fidelity prototype. Subsequently, we transformed that prototype into a fully functional, production-ready application, with a strong emphasis on usability, performance, and scalability. Additionally, we conducted testing to assess the soundness of the system's business logic, its performance, and its usability.

CCS Concepts

• **Human-centered computing** → **Information visualization**; *Ubiquitous and mobile computing systems and tools*; Social networking sites.

Keywords

Cultural tourism, Mobile App design, Literary tourism, Interactive mapping

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1 Introduction and Related Work

Over the past three decades, the tourism industry has grown steadily in revenue and participation [18], driven by innovations in transportation, multicultural societies, and evolving tourist behaviour [5]. Among these factors, the integration of digital technologies has been the most transformative, enabling more personalised and specialised tourism experiences [2]. Tourism has diversified into

sectors such as cultural, self-improvement, and rural tourism [11], with cultural tourism—and particularly *literary tourism*—offering visitors a deeper connection to a destination's culture [7]. Yet, few applications place literature at the centre of the experience, often limiting themselves to conventional content, such as simple guided routes with little cultural information [16] [4]. To fill this gap, we designed a mobile application that offers immersive literary routes with historical and multimedia content, supported by an interactive map. The first deployment of the application targets the Lisbon metropolitan area.

1.1 Mobile and Digital Tools for Cultural Tourism

Digital technologies have transformed cultural tourism, turning passive visitors into active participants who personalise their journeys [19]. Tools like augmented and virtual reality provide immersive experiences, while mobile apps and online platforms provide easy access to information, enabling tailored itineraries [3, 8, 13]. Social platforms allow tourists to share experiences, influencing others' travel decisions, and fostering community [6, 12]. Challenges include balancing technology with authentic experiences, bridging the digital divide, and ensuring data privacy [8, 14, 19].

Cultural tourists benefit from mobile applications that provide accessible, informative, and interactive tools throughout their journey [3]. Key features include detailed information on popular attractions, itinerary suggestions, context-aware notifications, social networking, and gamification. These apps not only support planning and navigation but also enrich the tourist experience by offering educational content and interactive storytelling. Such platforms can transform a simple visit into a deeply immersive and memorable cultural experience. Examples include *Smartify*¹, offering museum guides and tours and *Google Arts & Culture*², serving as a digital repository for artworks and cultural content.

When narrowing the scope to literary tourism applications, existing solutions are scarce and largely confined to specific regions. Examples include *dtour*³, which highlights literary sites in Otago, New Zealand; *Spot-Lit*⁴, a regional initiative supporting literary tourism in northern Europe; *Abracapp*, offering virtual tours and social features in Italy [17]; and *LitLong*, displaying geoparsed literature in



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¹<https://smartify.org/>

²<https://artsandculture.google.com/>

³<https://www.cityoflitterature.co.nz/a-literary-city/d-tour-App>

⁴<https://www.spot-lit.eu/>

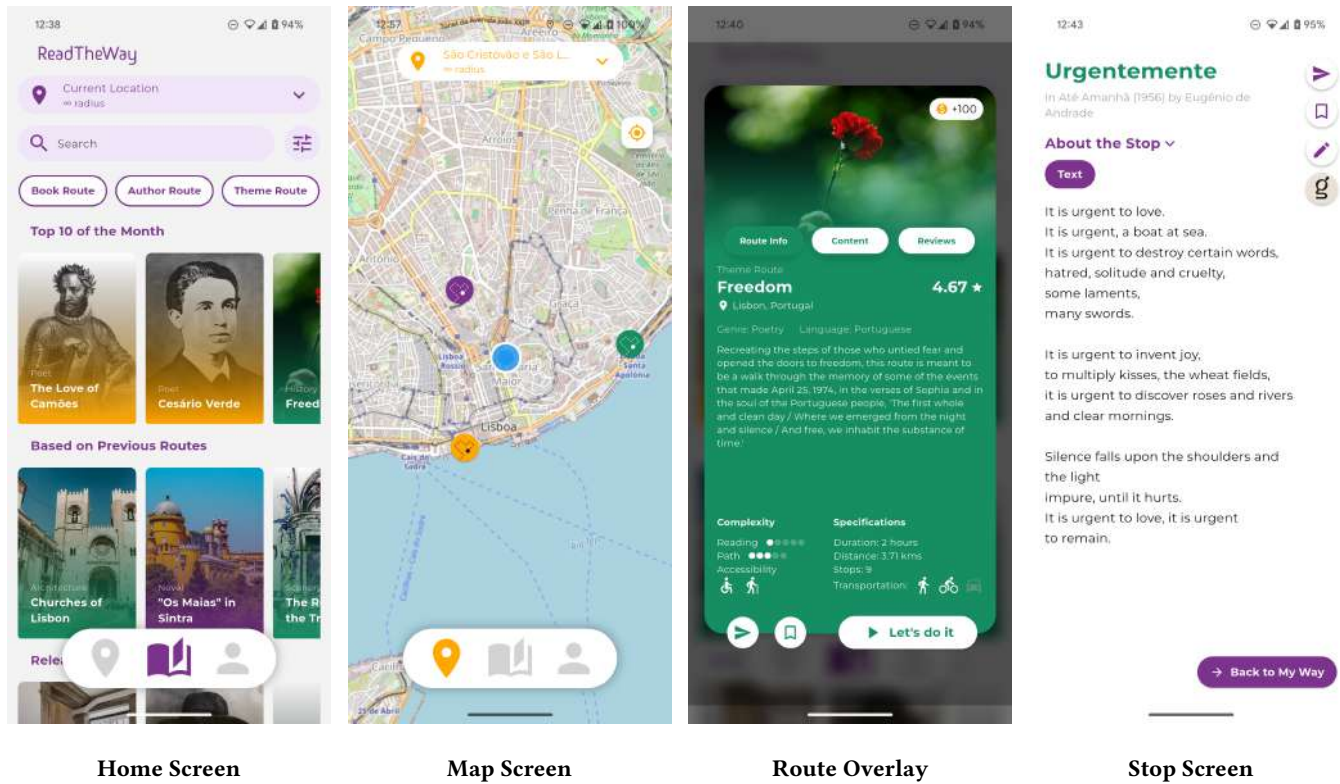


Figure 1: Key UI screens and features of the ReadTheWay mobile application that aims to promote literary tourism. The *Home Screen* allows for route discovery, the *Map Screen* 2D navigation, the *Route Overlay* detailed route information, and the *Stop Screen* information regarding each of the stops in the trajectory. This particular “Freedom” route follows key locations associated with the Carnation Revolution, one of the most important events in Portuguese history. It features poems and texts, many of them political protest works, towards a deeper understanding of the revolution’s cultural and historical significance.

Edinburgh [1]. Despite these examples, no existing widely available application combines literary content with interactive features such as mapping, gamification, or support for diverse media formats. This highlights a clear gap for applications like ReadTheWay, which aims to provide an immersive, literature-driven experience integrating multimedia and navigation.

2 System Design and Architecture

After extensive research on cultural tourism web and mobile applications, we identified five main categories of functional requirements: information provision, itinerary management, context awareness, social networking, and gamification. We argue that there has not previously been a tourism application centred specifically on literature that integrates all five functional areas. This integrated approach represents a key point of innovation in our project.

2.1 User Interviews and System Requirements

After identifying a market gap for a literature-centred tourism application, our focus shifted to understanding user behaviours, motivations, and contexts of use. Insights gathered from an online questionnaire with approximately 150 respondents who showed interest in a tourism App centred on literature, followed by in-depth

interviews with around 20 participants, revealed recurring patterns in how people combine travel, reading, and cultural exploration – primarily how most leverage travel as a way to connect with foreign cultures through museums, exhibitions, theatres, or historical landmarks. Their ages ranged from 19 to 50 years old, with most in their 20s; around two-thirds were female, nearly 90% had travel experience, and roughly half reported using travel Apps regularly.

These insights informed our *system requirements* (R). This research revealed that reading helps users immerse themselves in a place’s atmosphere and extend the travel experience beyond physical sightseeing, underscoring the need for a wide range of literary routes with significant cultural context (R1). Interest in lesser-known cultural contexts and personal author insights should lead to contextual and biographical content (R2). Diverse planning styles call for location-aware content delivery for dynamic discovery (R3), while frequent reading on the move and limited connectivity highlight the need for offline access (R4). Frustration with generic recommendations should motivate personalised content based on reading and travel interests (R5), and the desire to associate books with places and memories encourages the inclusion of features for saving and revisiting literary journeys (R6). Other requirements derived from our survey and interviews are omitted for brevity.

2.2 High-Level App Architecture

Figure 2 presents a high-level overview of the ReadTheWay platform. The system consists of a Flutter-based mobile application, a Spring Boot backend server, a PostgreSQL database, and several external services for mapping and media storage. The mobile App is organised into two main layers: the UI layer for information presentation and interaction, and the Data layer for business logic, data access, and persistence. The App communicates with the back-end to handle authentication, route fetching, and user data management. The back-end employs a traditional three-layer architecture: Controller, Service, and Repository, with a security module that manages JWT authentication. Persistent data is stored in a PostgreSQL database, while external APIs such as OpenStreetMap⁵ and HERE⁶ provide advanced mapping, routing, and public transit capabilities. Google Firebase Storage handles media assets to keep the database lightweight and efficient, with local caching implemented to reduce repeated network requests and improve responsiveness.

3 Prototyping and Development

3.1 Back-end Development

Our back-end was developed with Spring Boot (Java), providing a RESTful API for the Flutter frontend and handling data storage, user authentication, business logic, and integration with external APIs, including HERE Maps. The back-end follows a domain-oriented folder structure, with each core entity (Authors, Notes, Reviews, Routes, Shop, Stops, Users) having its corresponding folder containing the entity, DTOs, Controllers, Services, and Repositories. Security is enforced using Spring Security and JWTs. Passwords are stored in hashed and salted format using BCrypt, and login generates a 15-minute access token and a 30-day hashed refresh token. Rate limiting is applied to login and refresh endpoints to mitigate brute-force attacks. Client requests include timestamps and nonces to prevent replay attacks.

3.2 Front-end Development

The frontend was developed with Flutter, following a Model-View-ViewModel (MVVM) architecture. Navigation uses Flutter's Navigator API declaratively. Riverpod providers manage state reactively: *StateProviders* for simple, mutable state; *StreamProviders* for real-time updates; and services/viewmodels as providers for global access. Local storage with Hive caches routes, stops, and user data. When the client accesses the home page, it sends locally stored route IDs to receive only new or updated routes. Media assets are cached via Flutter Cache Manager, with automatic compression for large files. Given that certain routes may take users to areas with low connectivity, user requests are handled locally and placed in a *pending actions listener*. This ensures that all operations are stored and sent to the backend, in the correct order, once the App regains online connectivity. For example, if a user takes a note of a route while offline, the locally stored instance of the user is updated and saved, but the action is queued in the *pending actions listener*. As soon as network connectivity is restored and the server becomes available, these actions are automatically synchronised with the

backend. This ensures that users never lose their progress, even when switching between online and offline environments.

3.3 UI Design

The UI design was conceptualised to meet the system requirements described earlier. Figure 1 presents key UI screens of our application: the *Home Screen* for route discovery, the *Map Screen* for navigation, the *Route Overlay* for detailed route information, and the *Stop Screen* for viewing individual stop content. These screens illustrate the visual consistency and primary functionalities available to users. Because ReadTheWay aims to expand beyond Lisbon, our UI design choices were not influenced by any specific city. Instead, the App's visual identity is shaped by the type of literary content it presents. One example concerns the routes available in the App, which are divided into three categories: theme routes, author routes, and book routes. Each category features a distinct aesthetic, including different thumbnails and colour schemes, allowing users to immediately distinguish between genres.

4 Evaluation

The evaluation phase of our project comprehensively assessed functionality, efficiency, and usability. It comprised two complementary parts: performance testing and usability testing. In addition, iterative field testing and debugging throughout development provided real-world validation. The application was also reviewed by other members of our team, confirming that all functional, system, and design requirements were fully met. These team members tested several routes to validate the App's functionality and overall reliability. During these tests, we were able to navigate the routes using only the App's interface (i.e., without relying on any external navigation tools). The location marker and route *polylines* accurately corresponded to real-world positions, and all stops, texts, and associated content functioned as intended. Overall, these internal tests confirmed the App's usability and technical consistency in real-world conditions.

4.1 Performance Testing

Performance testing aimed to ensure that the application could run smoothly without placing excessive load on device resources. All tests were conducted on an Android device using Android Studio Profiler and Flutter DevTools, with each measurement repeated three times. CPU usage was monitored to evaluate processing efficiency. The application predominantly used small and mid cores, while the high-performance core was rarely activated, indicating that the App executes operations efficiently without relying on high-power processing [15]. Memory consumption was tracked both as Resident Set Size (RSS) and Dart Heap usage, with RSS peaking at ≈ 800 MB and Dart Heap stabilising at ≈ 70 MB, reflecting efficient memory management and minimal fragmentation.

Battery consumption was also analysed, using average and cumulative power usage across system components. The main contributors were the display and CPU (Little and Mid cores), while GPU, memory, and other subsystems contributed modestly. Overall, the application's power usage was consistent with typical mobile applications and remained within acceptable limits (≈ 50 mWh). Storage usage was also evaluated. The installed application size

⁵<https://www.openstreetmap.org/>

⁶<https://www.here.com/>

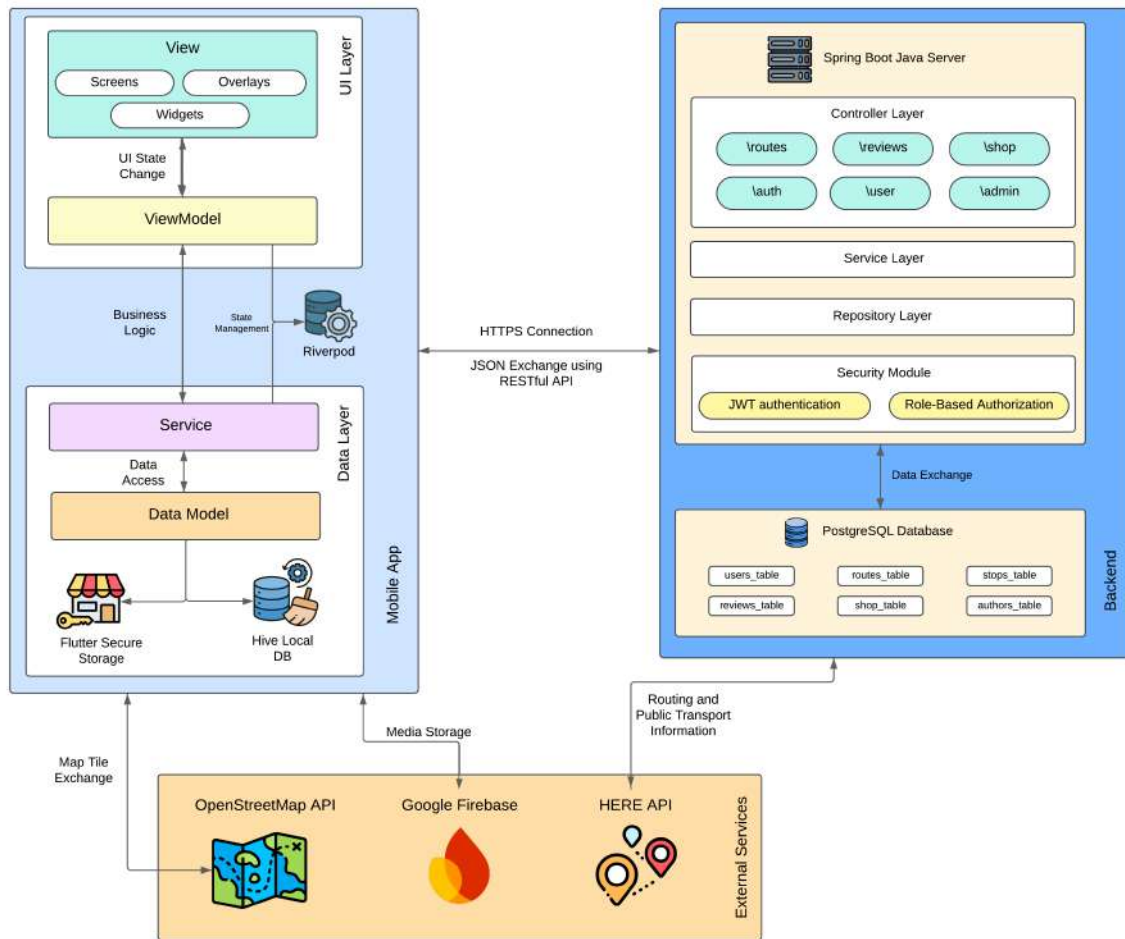


Figure 2: High-level diagram of the ReadTheWay architecture.

was approximately 40 MB, with caching consuming an additional 20 MB after extended use. This demonstrates that the application scales well and does not excessively consume storage over time. Finally, frame rendering performance was tracked to assess visual smoothness. The App achieved an average of 115 FPS, with only 3.62% of frames classified as *jank* (rendering times above 16 ms), which indicates that animations, scrolling, and transitions are fluid.

4.2 Usability Testing

4.2.1 Method. Usability testing evaluated the application’s interaction quality, user satisfaction, and overall experience. Twelve participants interacted with the fully functional application, allowing real-world insights into interface behaviour, network delays, and performance responsiveness. The twelve participants represented a diverse demographic profile. Ages ranged from 15 to 59 years, with most participants in their early twenties. Seven were identified as female and five as male. Prior experience with tourism applications varied, ranging from none to extensive. They were asked to

complete tasks including signing up (T1), changing profile pictures (T2), browsing routes (T3), taking notes (T4), saving stops (T5), and examining profile statistics (T6). During this process, participants followed a think-aloud protocol, verbalising their thoughts and reactions. After task completion, participants completed a structured questionnaire comprising demographic and background questions, the System Usability Scale (SUS) [10], the User Experience Questionnaire (UEQ) [9], and open-ended feedback.

4.2.2 Analysis. Our observations revealed that all tasks were completed successfully, demonstrating that the application’s overall workflow is intuitive. Minor issues were noted, including occasional unresponsive buttons when saving notes (T4: “The button to submit notes is unresponsive and it led me to click it multiple times”), inconsistent scrolling in certain screens (T9: “The stop screen scrolling should not cut the screen in half”), and the need for password confirmation during sign-up (T1: “I was unsure I typed the right password”). Participants also suggested that underlined annotations should persist after submission (T4: “I forgot what part

of the poem I had annotated”) and that previous and subsequent route segments should be displayed in grey on the map for better spatial orientation. Despite these points, participants consistently praised the application for its responsiveness, smooth transitions, vibrant design, and engaging combination of tourism content with literary features.

Quantitative analysis confirmed these findings. SUS scores yielded a mean of 84.58 ± 7.67 , indicating above-average usability within the “good to excellent” range. The UEQ analysis highlighted strengths in Attractiveness (2.33 ± 0.56) and Dependability (2.27 ± 0.43), with positive results also in Stimulation (2.19 ± 0.59) and Efficiency (1.94 ± 0.53). Perspicuity (1.54 ± 0.68) and Novelty (2.02 ± 0.48) were slightly lower but still positive, reflecting minor opportunities for refinement. Overall, the application was perceived as usable, engaging, and visually appealing. Based on participant feedback, minor improvements were implemented, such as loading indicators, improved scrolling behaviour, sign-up validation, annotation persistence, and map context.

4.3 Future Work

While the current version of the ReadTheWay application fulfils our initial objectives, future enhancements could improve its UX and system performance:

- **Extended functional scope:** integrating social media, augmented reality, or other gamification features, beyond the basic rewards system already present in the App;
- **Map cache optimisation:** implement clean-up to reduce RSS memory usage, especially on low-end devices;
- **More responsive map API tiles:** improving navigation responsiveness by switching from free OpenStreetMap tiles to higher-performance solutions such as Google Maps⁷, which offers better caching and infrastructure;
- **Improve server concurrency :** optimise server-side thread management, using bounded thread pools, and minimise contention through fine-grained and optimistic locking to handle concurrent requests efficiently.

5 Conclusion

The development of ReadTheWay was both a technical challenge and an exploration of how digital tools can create new cultural experiences. The key goal was building a clear, organised system. Our mobile App focuses on smooth, intuitive, and responsive interactions, while the back-end ensures reliable and secure access to routes, texts, and user data. Efficient client-server data flow, local storage, and offline access were prioritised to maintain usability even in real-world contexts with unstable connections. Performance, usability, and maintainability guided all development choices. The system was designed not only to work but also to adapt, supporting new features, more users, and evolving interactions with literature and place. ReadTheWay shows how thoughtfully applied technology can enrich our cultural experiences. We hope our work can inspire and serve as a blueprint to further efforts that promote reading, literature, and cultural tourism.

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⁷<https://mapsplatform.google.com/lp/maps-apis/>